

LESSON 12.1

APPLYING THE DISTRIBUTIVE PROPERTY WITH VARIABLES



LESSON INTRODUCTION

MA.8.AR.1.2 Apply Properties of Operations to multiply two linear expressions with rational coefficients.

LEARNING TARGETS

- I can write equivalent algebraic expressions by applying the distributive property.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.1.1 Apply Properties of Operations to add and subtract linear expressions with rational coefficients.

WORKING TOWARD

MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.

ESSENTIAL QUESTIONS

- How can you use the distributive property to write equivalent algebraic expressions?

LESSON NARRATIVE

In this lesson, students explore the distributive property using multiple representations. Students connect how the area model shows the distributive property being applied and engage with repeated practice to build fluency in using the distributive property to write equivalent algebraic expressions.

COMPONENT SUMMARY

12.1.1 WARM-UP (~5 MIN.)

Rate growth mindset statements

12.1.3 EXPLORATION (15-20 MIN.)

Represent areas of rectangles using area models

12.1.5 GUIDED PRACTICE (10 MIN.)

Generate equivalent algebraic expressions using the distributive property

12.1.2 REFRESHER (5 MIN.)

Apply exponent laws to generate equivalent algebraic expressions

12.1.4 BRING IT TOGETHER (10-15 MIN.)

Apply the distributive property of multiplication over addition

12.1.6 WRAP-UP (5 MIN.)

Use the distributive property to rewrite expressions

RESOURCES

GLOSSARY

Key Terms:

- Distributive Property of Multiplication over Addition

Additional Terms:

- N/A

MATERIALS

- (Optional) Markers or Colored Pencils

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not understand when working with area models that the sign stays with the term. For example, in 12.1.4 Bring It Together, question 2a, students may write 3 and x in the boxes along the top of the area model rather than 3 and -x.

THINGS TO CONSIDER

- Teacher-led discussions throughout the unit should include making connections to properties of operations (i.e., the commutative property, associative property, and distributive property).

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

12.1.3 Exploration provides several opportunities for students to engage in the Think-Pair-Share protocol. For example, when students are asked to write an expression that represents the sum of Rectangles A and B, students should review the diagrams independently, then pair to share what expression (or expressions) they feel represent the sum of the areas. This can open a conversation between students who have different expressions to determine if they are both equivalent.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

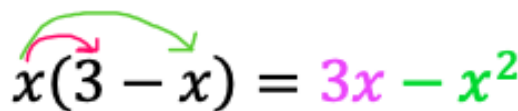
MA.K12.MTR.5.1 Patterns and Structure

Students make use of repeated patterns during 12.1.3 Exploration. Students have repeated opportunities to find the width, length, and area of different rectangles to be able to connect in the following component that the area model shows the distributive property being applied.

DIFFERENTIATE INSTRUCTION

Consider allowing students to use colored pencils or markers as a visual marker for students as they complete the area models and/or as they use the distributive property when given an expression.

Consider having students explain the process of applying the distributive property to an expression. A student could describe the process, while their partner is performing the steps described. This will help students build confidence and also be precise and specific in describing the process of applying the distributive property.



$$x(3 - x) = 3x - x^2$$

12.1.1 WARM-UP: GROWTH MINDSET

TEACHER MOVES

After students independently complete 12.1.1 Warm-Up, if time allows, consider having students share with a partner and discuss any statements on which they disagreed.



12.1.1 Warm-Up: Growth Mindset

- On a scale from 1 to 5, rate each of the following statements by circling the number that matches your belief.

1 = strongly disagree 2 = disagree 3 = neither/neutral 4 = agree 5 = strongly agree

a. Intelligence is something you're born with.	1	2	3	4	5
b. I can get smarter if I work hard.	1	2	3	4	5
c. Some things are just too difficult for some people to learn.	1	2	3	4	5

- Which statement did you agree with most? ____

- Explain why you agree with it most.
Answers will vary.



12.1.2 Refresher: Multiplying with Exponents

Any variable, a , can be rewritten using the exponent 1, as a^1 . This is the identity exponent law.

1. Complete the table by applying the identity exponent law and product of powers law to each given expression. The first row has been started.

Expression	Equivalent Expression Using Identity Exponent Law	Equivalent Expression Using Product of Powers Law
$d \cdot d$	$d^1 \cdot d^1$	d^2
$(4x)(x)$	$4x^1 \cdot x^1$	$4x^2$
$(-3f)(0.5f)$	$-3f^1 \cdot 0.5f^1$	$-1.5f^2$
$5c \cdot b \cdot 2c$	$5c^1 \cdot b^1 \cdot 2c^1$	$10bc^2$

2. Complete the statement.

Any variable, a , multiplied by itself can be rewritten as a^2 .

12.1.2 REFRESHER: MULTIPLYING WITH EXPONENTS

DIFFERENTIATE INSTRUCTION

Determine if your students need the support of all or part of 12.1.2 Refresher.

12.1.3 EXPLORATION: AREAS OF RECTANGLES

TEACHER MOVES

12.1.3 Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group 12.1.4 Bring It Together. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

TEACHER MOVES

There are multiple correct answers that should be accepted for question 1a.

$$3 \cdot x + 3 \cdot 6$$

$$3x + 3 \cdot 6$$

$$3x + 18$$

Students could also switch the order of the rectangle areas and be correct.



12.1.3 Exploration: Areas of Rectangles

Work with your partner to complete the following.

- Two rectangles are shown. The length and width of Rectangle B are 6 and 3. The width of Rectangle A is also 3, but its length is unknown, so it is labeled x .

Rectangle A	Rectangle B

- Write an expression to represent the sum of the areas of Rectangles A and B.

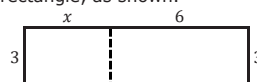
Answers will vary.

$$3 \cdot x + 3 \cdot 6$$

$$3x + 18$$

$$3x + 3 \cdot 6$$

- Rectangles A and B can be composed into one larger rectangle, as shown.



Complete the statement.

The width of the new rectangle is

length is

- ☐ 3.
- ☐ $(3 + x)$.
- ☐ 6.
- ☒ $(x + 6)$.

- ☒ 3
- ☐ $3x$
- ☐ 6
- ☐ 18

and the

- c. Tajae and Elyse wrote different expressions to represent the area of the larger rectangle.

Tajae	Elyse
$3(x) + 3(6)$	$3(x + 6)$

Discuss with your partner whether Tajae, Elyse, or both students wrote a correct expression of the area.

Answers will vary. Both students are correct.

- d. Write another expression to represent the area of the larger rectangle.

Answers will vary. $3x + 18$ or any equivalent expression.

2. For each rectangle, complete the table.
In the first two columns, write an expression to represent the width and length of the rectangle. Then, in the next two columns, express the area of the rectangle in two different ways using either multiplication or addition.

Rectangle A		2 3 6 x	
Width	Length	Area (width × length)	Equivalent Area Expression
$\frac{2}{3}$	$(6 + x)$	$\frac{2}{3} (6 + x)$	$4 + \frac{2}{3}x$

12.1.3 EXPLORATION: AREAS OF RECTANGLES (CONTINUED)

TEACHER MOVES

Think-Pair-Share

Students are only discussing here, not necessarily writing a response. Consider engaging students in a Think-Pair-Share protocol for question 1c.

Listen to student discussion to determine if the following connections are being made:

- Tajae’s expression is correct because it represents the sum of the areas of both smaller rectangles that compose the larger rectangle.
- Elyse’s expression is correct because it represents the product of the length and width of the larger rectangle.
- Both expressions are equivalent to each other and to the area of the larger rectangle.

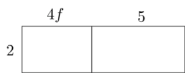
TEACHER MOVES

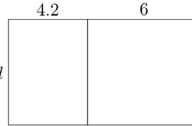
There are multiple correct answers for question 1d. Many students will likely finish multiplying to write $3x + 18$, but any equivalent expression would be correct.

12.1.3 EXPLORATION: AREAS OF RECTANGLES (CONTINUED)

ENRICHMENT

- Can we combine the terms $7y$ and y^2 ? Why or why not?
- How do we determine when we can combine terms?

Rectangle B			
Width	Length	Area (width × length)	Equivalent Area Expression
2	$(4f + 5)$	$2(4f + 5)$	$8f + 10$

Rectangle C			
Width	Length	Area (width × length)	Equivalent Area Expression
d	$(4.2 + 6)$	$d(4.2 + 6)$	$10.2d$

Rectangle D			
Width	Length	Area (width × length)	Equivalent Area Expression
y	$(7 + y)$	$y(7 + y)$	$7y + y^2$



12.1.4 Bring It Together: Distributive Property

The area models used throughout the Exploration demonstrate the **distributive property of multiplication over addition**.



Distributive Property of Multiplication Over Addition

$$a(b + c) = (a \times b) + (a \times c)$$

The factor outside of parentheses is “distributed” to each part of the factor within a set of parentheses. With the distributive property, multiplication is performed over (before) addition or subtraction.

1. Discuss with your partner how multiplication being performed before addition or subtraction relates to the order of operations.
2. Consider the example below that shows the application of the distributive property to multiply.

$$4(3 - x) = 12 - 4x$$

- a. Complete the area model to represent the same expressions by filling in the blanks for each factor.

	3	-x
4	12	-4x

- b. Discuss with your partner how the area model shows the distributive property.

- c. In the product $12 - 4x$, explain why the constant term is positive and the variable term is negative.
Answers will vary. The constant term is positive and when you multiply by 4, it stays positive. The variable term is negative and when you multiply by 4 it stays negative.

- d. Zion says the product could also be written as $-4x + 12$. Explain whether or not this is true.
Answers will vary. True; you can apply the commutative property to change the order of the sum $12 - 4x$ and rewrite it as $(12) + (-4x)$, or you can apply the commutative property to rewrite it as $-4x + 12$.

3. The distributive property can also be used to multiply algebraic expressions where variables are included in both factors. An example is shown.

$$x(3 - x) = 3x - x^2$$

Complete the statements.

The product of $x \cdot 3$ is $3x$. The exponent of x in this

term is as explained by the $3x$.
 $-x^2$.
 identity exponent.
 product of powers.

The product of $x \cdot (-x)$ is $3x$. The exponent of this

term is as explained by the $3x$.
 $-x^2$.
 identity exponent.
 product of powers.

Therefore, the factors of $x(3 - x)$ is $3x - x^2$, which

is determined by applying the distributive property to

multiply the factors x and $(3 - x)$.
 product

12.1.4 BRING IT TOGETHER: DISTRIBUTIVE PROPERTY

TEACHER MOVES

Question 2a, notice that the sign of each term stays with the term. Some students may write 3 and x in the boxes along the top of the area model rather than 3 and -x.

TEACHER MOVES

Question 2b, students should make connections between the two representations of multiplication. Both show the factor 4 being multiplied by both terms in the factor $(3 - x)$. It's important that students recognize $(3 - x)$ as a single entity with two parts rather than thinking both 3 and -x are factors because they are not. The factor being multiplied by 4 is the quantity $(3 - x)$.

TEACHER MOVES

Make connections to properties of operations throughout the unit. For example, $-4x + 12$ is equivalent to $12 - 4x$ because of the commutative property.

Students who understand that the commutative property only applies to multiplication and addition may be confused by this because they may think of $12 - 4x$ as subtraction.

$12 - 4x$ can be rewritten as $12 + (-4x)$ using the existence of additive inverse. Understanding that the terms 12 and $-4x$ are being combined (or added) in this expression will set students up to be better ready for operations with polynomials in Algebra 1. In algebraic expressions such as these, each term can be seen as either positive or negative with all being combined (or added) together. Therefore, the commutative property can be applied to change the order of the terms being added as long as the signs with each term remain the same.

12.1.5 GUIDED PRACTICE: DISTRIBUTING TERMS WITH VARIABLES

TEACHER MOVES

12.1.5 Guided Practice: Distributing Terms with Variables

DIFFERENTIATE INSTRUCTION

Encourage students to draw area models to represent the given expressions. Area models provide a visual representation of the expressions and can aid students as they distribute the factor to each term in parentheses. Do not make this a requirement, instead allow students to use it as a support.



12.1.5 Guided Practice: Distributing Terms with Variables

1. Generate an equivalent algebraic expression for each given expression by applying the distributive property.

a. $f(f - 6)$
 $f^2 - 6f$

b. $(0.2g + 1)5g$
 $g^2 + 5g$

c. $\frac{3}{2}c\left(2c + \frac{3}{2}\right)$
 $3c^2 + \frac{9}{4}c$

d. $-4x(x - 7)$
 $-4x^2 + 28x$

2. What is another equivalent expression for $-4x(x - 7)$ using the commutative property?
Answers will vary. Some students may rewrite the given multiplication problem as $(x - 7) \cdot (-4x)$, or may apply the commutative after multiplying. $28 - 4x^2$.

**12.1.6 Wrap-Up: Ticket Out the Door**

Rewrite each expression using the distributive property.

1. $(8 + 4.5z)(0.25z)$
 $2z + 1.125z^2$

2. $-15y\left(\frac{2}{5}y - \frac{1}{3}\right)$
 $-6y^2 + 5y$

12.1.6 WRAP-UP: TICKET OUT THE DOOR**DIFFERENTIATE INSTRUCTION**

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.